Cover catalog update for benchmark fixtures

2026-10-21 · Versioned metadata report · Geometry is not transfer advantage

Three synthetic cofactor models and five exact generator representations extend the catalog to 2,575 map rows, 2,573 unique cover models/Jacobian contexts, 982 exact EC1/curve/map triples and 510 declared-field contexts. The previous 2,572-row report and its evidence remain unchanged. Registry, covers and graph are frozen in commit 9be2f56ac440b5eeedf1438094a59e8795d66936.

Catalog-wide profile assessment (2026-10-21)

Exact current catalog
2,575 map rows / 2,573 cover models / 982 EC1 representations
all rows classified; classification is not blanket support

Map assessments: 2,575 / 2,575 exact rows

Profile	SUPPORTED	Explicit boundary
165 binary g3, degree 3	construction + geometry one-infinity Jacobian typed point-class pull/push	all-element oracle: UNKNOWN catalog arbitrary Jac(H) norm: REFUTED as current API fixed GF(8) fixture is outside catalog
1,056 odd prime direct	construction + geometry	two-infinity maps: UNKNOWN one-infinity use: REFUTED
30 odd prime normalized	construction + geometry degree-one normalization	coordinate commutation + balanced maps: UNKNOWN
1,324 GF(p^k)	same-declared-field construction + geometry	proper-subfield descent + balanced maps: UNKNOWN

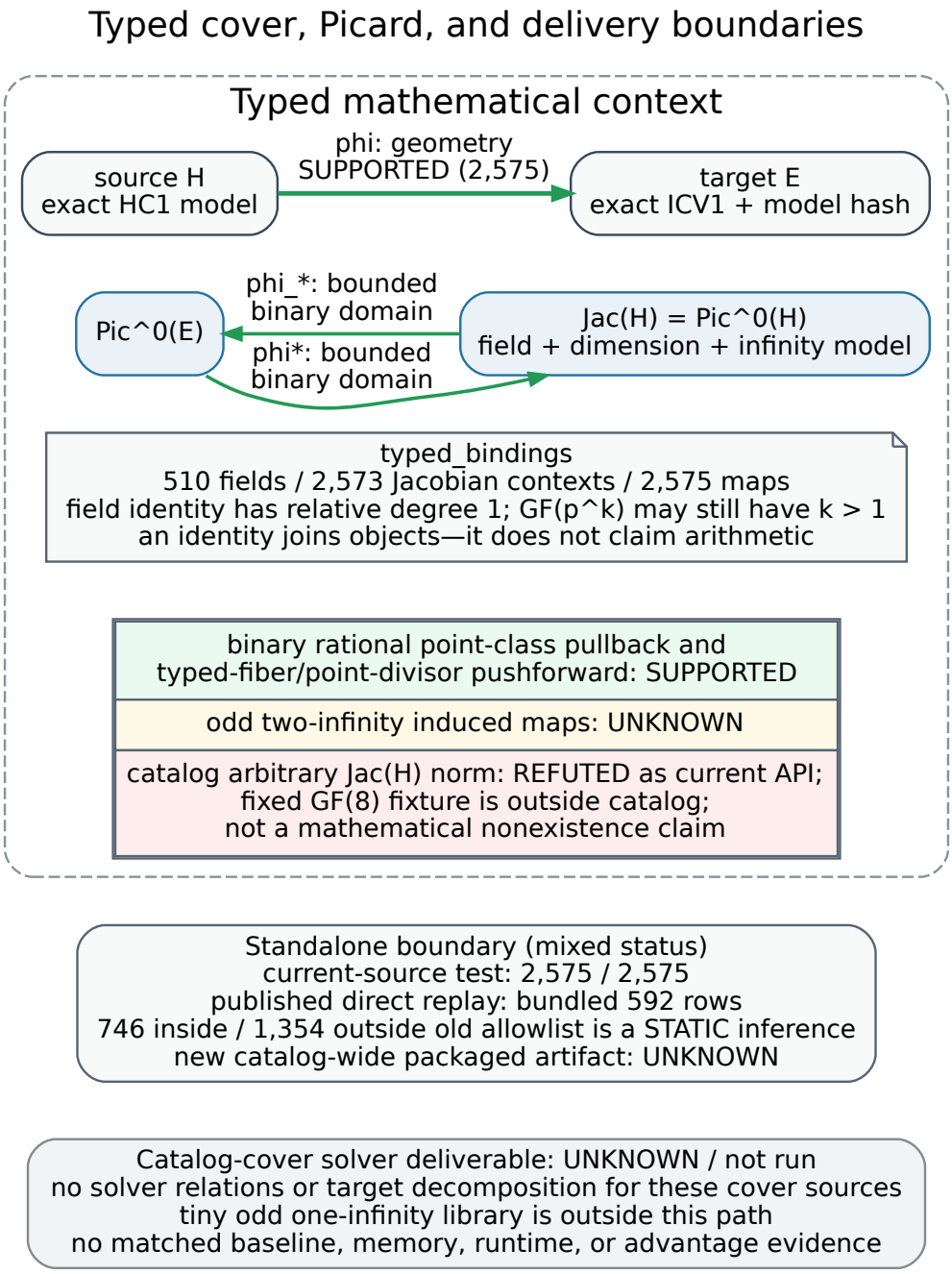
Subgroup assessments: 982 / 982 exact EC1 / curve / map triples

Profile	SUPPORTED	UNKNOWN unless overlaid
140 binary	identity, gcd(3,r), G on E, [r]G=O pullback, annihilation, composition, degree-inverse recovery of tested G	exact ord(G)=r; additivity; all-element injectivity; image order; arbitrary-result recovery
exact binary overlay	icv1-f2m7-t13-616700dd generator + image order 29	only that EC1/map triple
842 odd	identity + gcd(2,r)	generator/order evidence; every induced-map claim
exact odd overlay	icv1-fp10-t5-192cb216 map geometry + order-823 identity	no subgroup preservation follows

SUPPORTED on labeled domain	UNKNOWN / conditional	REFUTED implementation claim	MIXED: inline labels govern

The three added models have p=65519 and cofactors 2, 4 and 8. The other two new representations use [2]G and [7]G on the existing 16-bit benchmark model. They join the direct odd-prime quadratic geometry profile; no claim status or implementation domain is widened. README.md lists their exact identities.

Typed bindings and verification



Native generation and replay verify all 2,575 catalog geometry rows with zero invalid/unsupported inputs. Eleven native checker tests pass, including the entire catalog replay. The standalone-constructor test checks every current catalog model. The native 2,751-row trait rebuild matches the guarded incremental update byte for byte. See the study's verification logs.

The assessment retains its supported/refuted/unknown/not-applicable matrix. Geometry does not establish catalog-general induced Jacobian maps, an arbitrary norm, exact source subgroup order or a cover-integrated IC solver. The known-log ECDLP benchmark works directly on E and uses no cover transfer. This catalog update makes no performance claim. The older packaged-artifact receipt remains evidence only for its bundled 592 rows and historical allowlist comparison.

Sources: this README and graph sources; docs/curves/registry.json, covers.json, cover-links.yaml and cover-transfer-assessments.json; the live independent cover verifier and the study verification logs. The previous report is preserved at research/cover_transfer_assessment_20261013/README.md.